

Open Design Disclosure

LegStyle Prosthetic Covers

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Purpose and intent

I am an amputee who has spent many years designing and making prosthetic leg covers, initially for myself and later for others. My motivation has always been practical and personal: prosthetic covers should be lightweight, comfortable, durable, and expressive enough to fit naturally into everyday life.

Over time, elements of the LegStyle design approach have been shared through real-world use, collaboration with clinicians, and commercially supplied products. Rather than attempting to restrict access to those ideas, this document openly records the core design concepts used in LegStyle covers so they are clearly and publicly known.

This disclosure is made in the spirit of openness and progress. People deserve better prosthetic covers, and improving outcomes in this space benefits from shared understanding, care in execution, and trusted relationships.

If you are interested in working with someone who lives with a prosthetic limb, has been making covers for many years, and has built a reputation for practical, good-looking solutions, you are welcome to get in touch.

Overview

LegStyle covers are removable external covers intended for use on lower-limb prostheses. The covers are primarily cosmetic but also provide physical protection to the prosthetic limb while remaining lightweight, flexible, and suitable for everyday use, including exposure to water and general environmental conditions.

The design prioritises:

- installation and removal without removing the prosthetic limb from the user
- ease of installation and removal,
- secure attachment without permanent modification to the prosthesis,
- low mass and comfort during wear,

- adaptability to a wide range of socket geometries,
 - and durability in real-world conditions.
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Structure and materials

In one implementation, the cover comprises a shaped shell formed from a flexible or semi-rigid polymer material. The shell retains a defined external form while allowing controlled flex to accommodate movement and donning or doffing.

Preferred wall thicknesses may range approximately from **0.9 millimetres to 3 millimetres**. This range allows the cover to be:

- cut or trimmed where required,
- locally reinforced,
- stitched or sewn through if desired,
- repaired or modified without compromising overall function.

This construction results in a cover that is materially lighter than traditional rigid cosmetic covers while remaining robust and practical.

Materials may include elastomeric or thermoplastic polymers suitable for shaping or additive manufacturing. TPU is a common example but is not required.

Upper attachment and retention

The cover is configured to attach near an upper region adjacent to the prosthetic socket using **non-permanent attachment methods** that avoid rigid mechanical fixation.

In one preferred implementation, retention is achieved using a **hook-and-loop interface**, where:

- a hook material is attached to or integrated with the inner surface of the cover near its upper edge, and
- a corresponding loop material is associated with the prosthetic socket.

The loop material may be attached to the socket via a compliant interface layer. One example includes a laminate with:

- a rubberised or neoprene-like surface facing the socket, and
- a loop surface facing outward.

The hook-and-loop connection may be engaged under light stretch or tension to create secure retention while allowing repeated removal and minor geometric variation.

In an alternative implementation, a **stretchable loop tape** may be permanently adhered to the socket, allowing direct engagement with hook material on the cover.

These approaches avoid tools, allow repeated installation and removal, and do not require permanent modification of the prosthesis.

Lower retention and stiffness feature

The cover may include a **step feature or cut-out** at a lower region of the limb. This geometry:

- assists axial retention,
- provides local stiffness,
- and improves positional stability during movement.

The step feature may also function as a structural or visual transition region within the cover.

Access openings and closures

To enable fitting over complex prosthetic geometries, the cover may include an elongate opening extending along part of its length.

In one common implementation, this opening is closed using a **zip fastener**, preferably positioned along a less visually prominent region such as the inseam. The closure allows:

- opening for donning,
- secure closure once positioned,
- and reopening for removal or replacement.

Alternative closure mechanisms may be used provided they allow repeated use without tools and without permanent modification of the prosthesis.

Reinforcement and structural variation

LegStyle covers may incorporate **integrated or standalone reinforcement structures** in selected regions to provide localised stiffness, shape retention, and protection.

In one implementation, the reinforcement structure is hollow or partially hollow and conforms to both the internal geometry of the cover and the underlying prosthetic limb. This structure behaves in a foam-like manner while remaining **non-absorbent**, enabling:

- resistance to water absorption,
- easy cleaning,
- rapid drying,
- and suitability for wet or dirty environments.

Unlike conventional foams, these structures are formed from flexible or semi-rigid materials rather than absorbent cellular foams.

Reinforcement materials

Reinforcement structures may be formed from materials including but not limited to:

- soft or foamed TPU,
- flexible thermoplastic elastomers,
- semi-rigid TPU variants,
- flexible nylons,
- or other materials exhibiting elastic or semi-elastic behaviour.

Material stiffness may be selected to provide cushioning and conformity or increased structural support. TPU is not required; any material capable of flexing sufficiently for installation and removal while retaining structure is suitable.

Interaction with closures and additional retention mechanisms

Depending on geometry and material stiffness, some implementations may **reduce or eliminate the need for a closure mechanism** along the length of the cover.

For example:

- stiffer or semi-rigid covers may flex on and off the prosthesis without a zipper,
- retention may rely primarily on upper attachment features,
- additional rubberised or high-friction interface elements may assist positional stability.

In some implementations, additional retention may be provided through **rubberised strips sewn into the zipper region**, optionally including spacer features to increase friction. When zipped into place, these elements engage the socket geometry and local features such as valves or contours, allowing the cover to lock into position without requiring hook-and-loop attachment at the socket.

Both zipped and non-zipped configurations are explicitly supported by the disclosed design approach.

Fit and customisation

Cover geometry may be based on:

- repeatable base designs,
- measurement-derived sizing,
- or fully custom digital models of individual prosthetic limbs.

The design approach supports both repeat production and highly customised implementations.

Variations

The concepts described here are not limited to any specific material, geometry, closure type, or manufacturing method. Variations that achieve similar functional outcomes in attachment, retention, flexibility, usability, and durability are considered part of the disclosed design approach.

Disclosure history and timeline (non-exhaustive)

Development and disclosure of the concepts described above occurred progressively through real-world use, communication, and supplied products. The following timeline is provided for transparency and record-keeping and is not intended to limit scope.

- **March 2025**
Non-permanent socket attachment concepts communicated via email.
- **July 2025**
Localised stiffening regions between the cover and prosthetic hardware disclosed to improve durability and reduce wear.
- **24 August 2025**
Distal step feature and cut-out disclosed for retention and stability.
Reinforcement regions sewn into selected areas disclosed, including reinforcement to protect covers from protruding hardware such as pin-lock mechanisms.

Reinforcement placement described as position-selectable and optionally aligned with closure regions to minimise stitching.

- **24 August 2025 (documented images)**

Images exist documenting hook-and-loop-based socket attachment methods. Some disclosure occurred in person rather than through written correspondence.

- **7 December 2025**

Additional retention mechanism for use over a knee sleeve disclosed. This includes rubberised strips sewn into the zipper region, optionally incorporating spacers to increase friction. When zipped, engagement with socket geometry and local features enables retention without hook-and-loop attachment at the socket.

Additional features disclosed in this context include:

- sewing reinforcement regions into selected locations,
- combining reinforcement with zipper regions,
- sewing branding or identification labels inline with zipper regions for permanent attachment.

Earlier informal disclosures and real-world use also occurred prior to these dates.

Closing note

This disclosure documents design concepts already in real-world use and is intended to prevent accidental reinvention or restriction of ideas that should remain available to the prosthetic community.

LegStyle will continue to focus on execution, durability, fit, and collaboration with prosthetic users and clinicians. Progress in this space comes from care, experience, and doing the work well.